From recoil motion to companion access: controlled benchmarks and a cluster-snapshot comparison

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We examine the approximations connecting a retained black-hole remnant's motion to its access to potential companions. A sequence of controlled experiments separates ballistic return, density exposure, velocity-weighted close passages, and the transfer of assumed host and target distributions to an evolved cluster snapshot. In a static Plummer potential, a first-return-then-central prescription misses early exposure and eventually overestimates it, while exposure per bound excursion remains finite as the return time diverges near escape. For a specified dilute binary population, density exposure overestimates normalized passage counts in the selected higher-kick, focusing-dominated cases; the discrepancy follows from the relative-velocity dependence of the encounter kernel. An external cluster snapshot contains too few binary black holes to calibrate a continuous companion distribution, and matching its total mass and half-mass radius with a Plummer model does not reproduce its frozen-potential return times. Independent numerical formulations, analytical limits, and full-shell comparisons verify the stated calculations. Together, these results identify distinct approximation errors and calibration gaps. They support conditional benchmarks for companion accessibility, without establishing live recoil trajectories, companion-acquisition probabilities, repeated-merger rates, or observational formation-channel constraints.

I. INTRODUCTION

A retained merger remnant need not remain in the dense part of its host. Another merger requires more than retention: the remnant must reach suitable companions, acquire or enter a binary, and undergo subsequent evolution. Published work already treats recoil displacement and frictional recentering [1], host evolution and pairing [2], and successive mergers with explicit few-body encounters [3]. Black-hole concentration through mass segregation is also established [4]. The question here is narrower: which approximations preserve the information needed to estimate access to a specified target population?

We examine successive links in that calculation. E001 establishes the ballistic return of a centrally kicked test particle in the standard Plummer potential [5]. E002 compares continuous density exposure with two central-residence prescriptions. E003 supplies an explicit tracer-binary population and cross section, testing whether normalized density exposure represents normalized close-passage counts. E004 separately audits the target population and host potential of one evolved Cluster Monte Carlo (CMC) snapshot [6, 7]. The experiment identifiers connect the manuscript to independently versioned configurations and result records.

These steps do not constitute a calibrated end-to-end merger model. E003 is conditional on an imposed bath; E004 does not calculate passage counts in the CMC population or evolve a kicked remnant in a live cluster. Its host and physical kicks differ from E003's. We therefore neither combine their errors into a merger-rate correction nor interpret the snapshot as astrophysical validation of E003.

Our ballistic center crossing is distinct from frictional recentering, so the calculations do not reproduce or falsify the return prescription of Ref. [1]. Within the fixed one-parameter Plummer family, an exact return time and the governing equations can identify the orbit; the error lies in replacing that orbit by an assumed residence history. The gravitational-wave connection is to the environment of a possible merger remnant. No observed signals, radiation waveforms, or merger outcomes are analyzed.

II. MODEL AND NUMERICAL METHOD

With host mass M, scale radius b, and gravitational constant G, define

$$\Phi(r) = -\frac{GM}{\sqrt{r^2 + b^2}}, \tag{1}$$

$$t_d = \sqrt{\frac{b^3}{GM}}, \qquad \eta = \frac{v_k}{\sqrt{2GM/b}}. \tag{2}$$

Let x be signed displacement in units of b and let time be measured in t_d. The initial-value problem is

$$\ddot{x} = -\frac{x}{(1 + x^2)^{3/2}}, \tag{3}$$

$$x(0) = 0, \qquad \dot{x}(0) = \sqrt{2}\eta, \tag{4}$$

with conserved dimensionless specific energy $E = \eta^2 - 1$. We restrict attention to $0 \le \eta < 1$. For a nonzero kick, P is the interval between successive center crossings. Signed position has period $2P$; radius and density have period P. The exactly stationary case has no return event and is treated separately.

A. First central return: experiment E001

Energy conservation yields the apocenter

$$x_a = \frac{\eta\sqrt{2 - \eta^2}}{1 - \eta^2} \tag{5}$$

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and return time

$$P = 2\int_0^{x_a} \frac{dx}{\sqrt{2[(1+x^2)^{-1/2} - (1-\eta^2)]}}. \tag{6}$$

Set $x = x_a \sin\theta$, $s = \sqrt{1 + x_a^2 \sin^2\theta}$, and $s_a = (1-\eta^2)^{-1}$. The endpoint singularity is removed:

$$\frac{dt}{d\theta} = \sqrt{\frac{s s_a (s + s_a)}{2}}, \tag{7}$$

$$P = 2\int_0^{\pi/2} \frac{dt}{d\theta}\, d\theta. \tag{8}$$

B. Residence and exposure: experiment E002

For horizon H and operational sphere radius cb, define

$$C(H, c) = \int_0^H \mathbf{1}_{|x(t)|<c}\, dt, \tag{9}$$

$$I(H) = \int_0^H (1 + x(t)^2)^{-5/2}\, dt. \tag{10}$$

Here C/H is a residence fraction, while I/H is the mean host density relative to its central value. Neither is an encounter probability. The comparisons for nonzero kicks are

$$I_{\text{central}}(H) = H, \tag{11}$$

$$I_{\text{wait}}(H) = \max(H - P, 0); \tag{12}$$

both equal H for a stationary object.

The production calculation uses energy quadrature with absolute and relative tolerances of 10^{-11}. Exposure integrals are split at radii 0.01, 0.1, 1, 10, and 100 when these lie on the orbit. This resolves narrow central passages at large apocenter. Complete periods accumulate exactly; the partial period is recovered by a bracketed inversion of outgoing travel time. Core crossing times are found by quadrature, not by a time-sampled indicator.

III. FIRST-RETURN RESULTS

E001 tabulates nine kick fractions from 0.1 to 0.999. The plotted curve extends to 0.9999. Table I lists representative results, and Fig. 1 compares the numerical curve with its asymptotic limits. All lengths and times use the units in Sec. II.

The small-amplitude equation is harmonic, so $P \to \pi$ as $\eta \to 0^+$. At exactly zero kick, the particle instead remains stationary. Near escape,

$$P \sim \frac{\pi}{\sqrt{2}}(1-\eta^2)^{-3/2}. \tag{13}$$

At $\eta = 0.99$, the apocenter is about $50b$ and the excursion takes about $793 t_d$. Such large excursions also expose the limitations of the isolated potential: tides, external structure, or host evolution could materially alter a real remnant's trajectory. The tabulated values are conditional on the stated ideal model.

TABLE I. Representative first-return results (E001). The apocenter is in units of b; P is in units of t_d.

η	x_a	P
0.5	0.881917	4.395561
0.8	2.591534	11.371439
0.9	5.167285	27.972769
0.95	10.207544	74.156668
0.99	50.241305	792.551744
0.999	500.249126	24856.344701

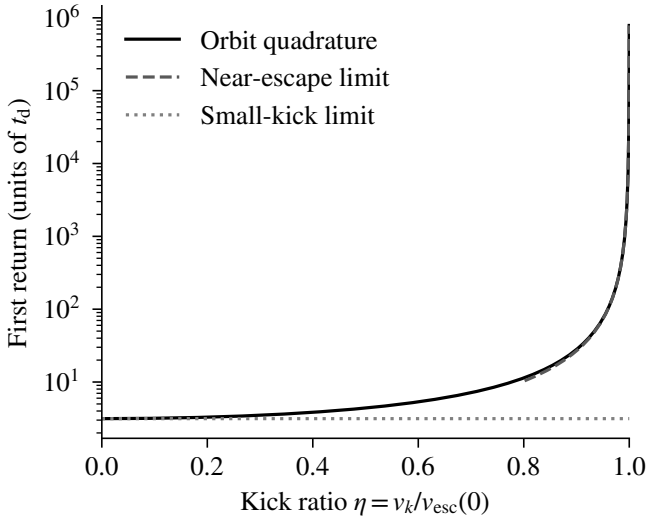


FIG. 1. First central-return time from energy quadrature, with small-kick and near-escape limits. Binding alone does not imply a short excursion. The return is a crossing, not dissipative settling.

IV. RESIDENCE AND EXPOSURE RESULTS

E002 uses 11 kicks from 0 to 0.999 and spheres with $c = 0.1, 0.3, 1$. Seven fixed horizons span 1 to 1000 dynamical times. For the ten nonstationary kicks, a second grid uses $H/P = 0.1, 0.5, 1, 2, 10$. Fixed and period-scaled windows are distinct comparisons.

At $\eta = 0.99$, the time inside $r < b$ is 1.530574 dynamical times per radial period. The mean sampled density is approximately 0.130% of the central value, while the residence fraction inside this sphere is 0.193% (Table II). These percentages describe one ideal orbit, not an observed population.

At $H = P$, the wait prescription gives zero, whereas the orbit has accumulated $I(P) > 0$. At $H = 10P$ for $\eta = 0.99$, the orbit gives 10.292151 dynamical times of exposure and the wait prescription gives 7132.965700. The error changes sign (Fig. 2). More generally, as H grows, the wait prescription approaches unit mean density while the orbit approaches $I(P)/P < 1$ for every nonzero kick. It is therefore not uniformly conservative.

An orbit stays within a sphere when its apocenter does not

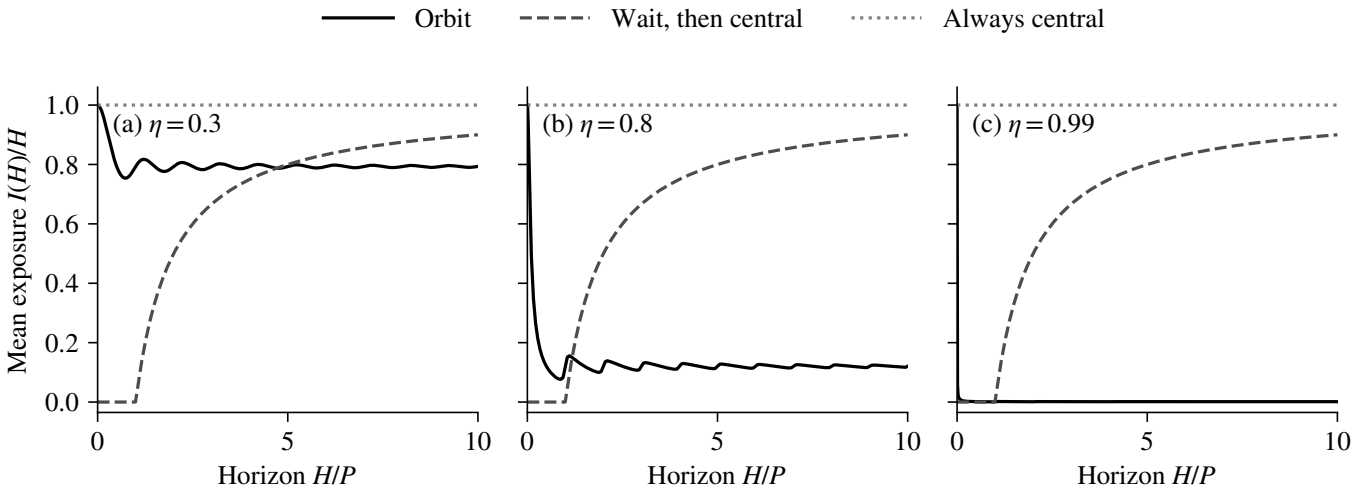


FIG. 2. Mean normalized density sampled up to each horizon for three kick fractions: (a) $\eta = 0.3$, (b) $\eta = 0.8$, and (c) $\eta = 0.99$. The first-return comparator suppresses all initial exposure and then assumes unit density permanently. The undamped orbit repeatedly departs from the center. Line styles distinguish the prescriptions independently of color.

TABLE II. Density exposure and residence over one radial period (E002). All entries are dimensionless; $C(P, 1)/P$ is the fraction of a period inside $r < b$.

η	$I(P)$	$I(P)/P$	$C(P, 1)/P$
0.3	2.774830	0.793277	1
0.8	1.369228	0.120409	0.174465
0.99	1.029215	0.00129861	0.0019312

exceed its radius:

$$\eta \le \sqrt{1 - \frac{1}{\sqrt{1 + c^2}}}. \tag{14}$$

For $c = 1$, the threshold is 0.541196. Equality touches the sphere boundary instantaneously and does not change an integrated residence fraction. Even an orbit entirely within the sphere samples a varying density; residence and density exposure are different quantities. Figure 3 shows the dependence on the operational sphere radius.

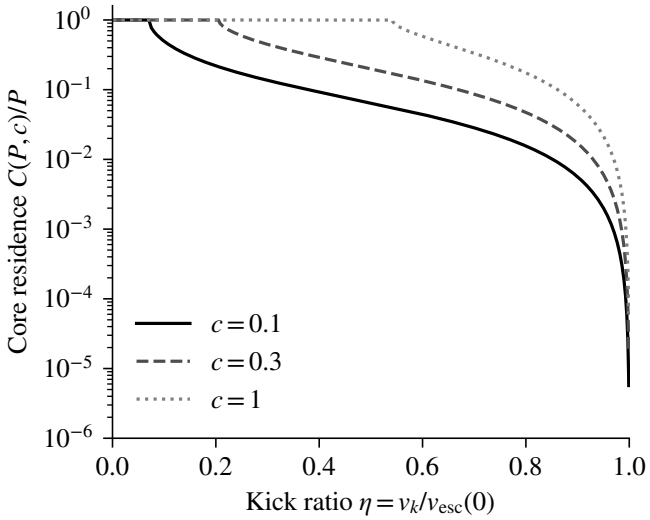


FIG. 3. Residence fraction per radial period for three operational sphere radii. Each curve equals unity below the threshold in Eq. (14) and decreases as bound kicks approach escape. The radii are model definitions, not measured cluster core radii.

V. FROM DENSITY EXPOSURE TO CLOSE PASSAGES: E003

A. Specified population and comparison

We set $M = 10^6\,M_\odot$, $b = 1\,\mathrm{pc}$, and remnant mass $m_\mathrm{r} = 60\,M_\odot$. An imposed population of $N_\mathrm{b} = 100$ tracer binary black holes has total mass $m_\mathrm{b} = 60\,M_\odot$ per binary and semi-major axis $a = 0.1$ AU. Its mass is 0.6% of the fixed host mass. The unsegregated target number density is

$$n_\mathrm{b}(r) = n_0 \widetilde{\rho}(x), \qquad n_0 = \frac{3N_\mathrm{b}}{4\pi b^3}, \quad \widetilde{\rho}(x) = (1 + x^2)^{-5/2}. \tag{15}$$

The target center-of-mass velocities follow the isotropic equilibrium Plummer distribution, not a measured black-hole subsystem.

All four cases, $\eta = 0, 0.3, 0.8, 0.9$, share $H = 10$, corresponding to $T = Ht_d = 0.149096\,\mathrm{Myr}$. They are admitted only if their apocenters lie within five Plummer half-mass radii, $6.52383b$. This operational screen is motivated by

Ref. [1]; it does not truncate the potential or target distribution. The E002 $\eta = 0.99$ example fails that screen and remains an isolated-model limit. A common elapsed time is essential: the higher-kick cases have not completed one return by $H = 10$, whereas the $\eta = 0.3$ case has completed about 2.86 radial periods.

A close passage is defined by two-body monopole pericenter $R = 1$ AU from the target binary center of mass. This is ten binary semimajor axes, a boundary around the binary's neighborhood rather than a cross section for exchange. For relative speed w before mutual focusing,

$$\Sigma(w) = \pi R^2 \left[1 + \frac{2G(m_{\rm r} + m_{\rm b})}{Rw^2}\right]. \tag{16}$$

The local two-body approximation neglects host tides during the approach and internal binary multipoles. It does not determine the outcome after entry.

B. Velocity weighting and the density approximation

Let u be target speed and v remnant speed in the host frame. At fixed radius, $y = u/v_{\rm esc}(r)$ has density

$$p(y) = \frac{512}{7\pi} y^2 (1 - y^2)^{7/2}, \qquad 0 \le y \le 1. \tag{17}$$

This follows from the isotropic Plummer distribution $f(E) \propto (-E)^{7/2}$ [8]. Isotropic angular averaging gives

$$\begin{aligned} \langle w \rangle_\Omega &= \max(v, u) + \frac{\min(v, u)^2}{3 \max(v, u)}, \\ \langle w^{-1} \rangle_\Omega &= \frac{1}{\max(v, u)}. \end{aligned} \tag{18}$$

The velocity-integrated zero-speed limit is finite. We evaluate the remaining speed moments with incomplete beta functions and check them by direct numerical angle and speed integration.

Define the local encounter kernel and passage rate by

$$\begin{aligned} K(r, v) &= \pi R^2 \left[\langle w \rangle + \frac{2G(m_{\rm r} + m_{\rm b})}{R} \langle w^{-1} \rangle\right], \\ \Gamma(r, v) &= n_{\rm b}(r) K(r, v), \\ \tau(T) &= \int_0^T \Gamma[r(t_{\rm phys}), v(t_{\rm phys})]\, dt_{\rm phys}. \end{aligned} \tag{19}$$

Here K has dimensions of volume per time, Γ inverse time, and τ is a dimensionless expected passage count along the unperturbed orbit. We do not turn τ into a merger probability or assume that the orbit remains unchanged after an actual encounter.

For implementation, use $V = \sqrt{GM/b}$ and dimensionless $t = t_{\rm phys}/t_d$. The geometric and focusing integrals are

$$\begin{aligned} J_g &= \int_0^H \widetilde{\rho} \langle w/V \rangle\, dt, \qquad J_f = \int_0^H \widetilde{\rho} \langle V/w \rangle\, dt, \\ \tau &= n_0 \pi R^2 b \left(J_g + \lambda J_f\right), \qquad \lambda = \frac{2(m_{\rm r} + m_{\rm b})b}{MR}. \end{aligned} \tag{20}$$

TABLE III. E003 at the common horizon $H = 10$. D and Q are normalized to the stationary control; $\delta = D/Q - 1$ is reported in percent. τ is an expected passage count, not an observed count or merger probability.

η	D	Q	δ (%)	τ
0	1.00000	1.00000	0.00	1.608×10^{-6}
0.3	0.78472	0.75156	4.41	1.209×10^{-6}
0.8	0.07626	0.04982	53.06	8.011×10^{-8}
0.9	0.05811	0.02927	98.55	4.706×10^{-8}

The length factor $b = Vt_d$ applies the time conversion once. In this configuration $\lambda = 49.5036$.

The stationary central control defines $Q = \tau(\eta, H)/\tau(0, H)$. The density proxy is $D = I(H)/H$. Their relationship is exact for the specified bath:

$$\frac{Q}{D} = \frac{\int_0^H \widetilde{\rho}[x(t)] K[r(t), v(t)]\, dt}{K(0, 0) \int_0^H \widetilde{\rho}[x(t)]\, dt}. \tag{21}$$

Thus the density approximation discards variation in the encounter kernel. It is accurate when the density-weighted mean of K is close to its stationary central value; residence alone does not enforce that condition.

C. Results and physical interpretation

The signed proxy error is $\delta = D/Q - 1$. Before execution, we declared $|\delta| > 0.10$ a failure of this benchmark's approximation screen. This is an operational tolerance, not a significance level or a requirement from gravitational-wave inference.

Table III and Fig. 4 show errors of +4.41%, +53.06%, and +98.55% at the three nonzero kicks. The two larger kicks fail the screen. At $\eta = 0.9$, density exposure is 5.81% of the control but passage count is 2.93%. The density-only approximation is nearly twice the conditional velocity-weighted count. The ordering across kicks does not reverse.

The geometric and focusing terms respond differently to motion. At $\eta = 0.9$, their separately normalized contributions are 0.1110 and 0.02867, respectively, compared with $D = 0.05811$. Higher relative speed increases geometric flux but reduces the inverse-speed contribution of focusing. Focusing supplies 97.23–99.19% of the nonzero-kick counts and controls the total discrepancy. Slow motion near apocenter does not compensate for the low tracer density there.

Absolute counts remain small: the stationary control has $\tau = 1.6080 \times 10^{-6}$ and the $\eta = 0.9$ case has $\tau = 4.7064 \times 10^{-8}$. These are conditional outputs for the imposed population and window, not an encounter-rate prediction for a calibrated cluster. A large fractional proxy error here does not establish a large merger-rate effect. The short window also does not prove that friction, diffusion, or host evolution would be negligible in a real system.

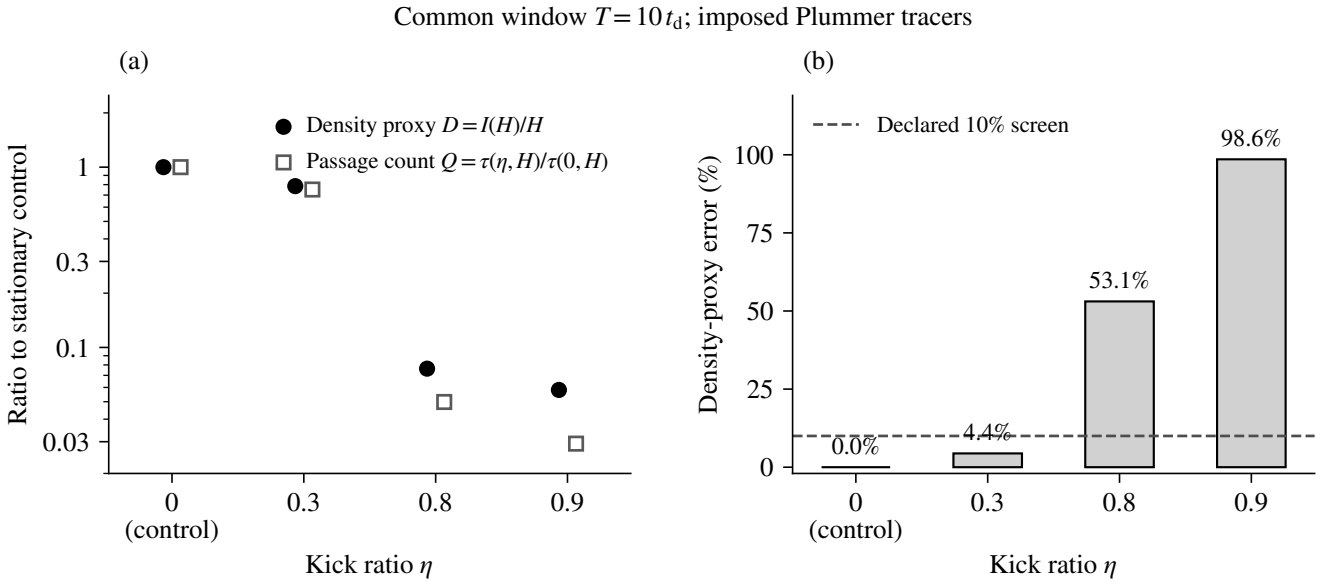


FIG. 4. E003: (a) normalized density exposure D and velocity-weighted passage count Q over the same elapsed time; (b) signed density-proxy error and the predeclared 10% screen. The central stationary case is the normalization control. Values are deterministic calculations, without sampling error bars.

VI. EXTERNAL POPULATION AND POTENTIAL AUDIT: E004

A. Reference state and comparison design

We use the full evolved example distributed with `cmctoolkit` [9]: model `N8-RV1-RG8-Z0.1`, snapshot `initial.snap0382.dat.gz`, at the pinned source revision identified in Ref. [7]. It was selected as an available evolved example before examining discrepancies. It is one simulated state, not an observational dataset or the full CMC catalog [10].

The age is 12.7323 Gyr, total system mass $M = 224633\,M_\odot$, and three-dimensional half-mass radius $r_h = 4.79771$ pc. Matching these last two quantities gives Plummer scale $b = r_h\sqrt{2^{2/3} - 1} = 3.67706$ pc. E004 uses these current properties, not initial conditions or E003's host parameters. The numerical kick fractions recur across experiments, but their physical speeds are not the same across hosts.

We separately count all systems, all binaries, single black holes, binaries containing at least one black hole, and binaries containing two black holes (BBHs). Stellar-type code 14 identifies black holes in this input convention. The declared minimum count of 30 screens whether continuous BBH-distribution calibration is supported; it is not a statistical sufficiency theorem. Descriptive radial and speed CDF distances carry no independent-sample significance interpretation.

TABLE IV. E004 inventory. Radius is the number-median system radius divided by the host's mass-weighted half-mass radius. Categories overlap: BBHs are a subset of BH-bearing binaries.

Population	N	median$(r)/r_h$
All systems	682,583	1.24677
All binaries	31,122	1.09910
Single BHs	21	0.04018
BH-bearing binaries	6	0.03851
BBHs	2	0.01014

B. What the companion sample supports

There are 682583 systems, 31122 binaries, and 29 black-hole objects. Only two systems are BBHs (Table IV). Their total masses are 26.672 and 29.821 $M_\odot$, and their semimajor axes span 0.23958–12.40947 AU. Their median radius is 0.04864 pc, or $0.01014\,r_h$. All binaries and all black holes would be inappropriate replacements for this specific target population.

These two centrally concentrated binaries do not calibrate E003's fixed-mass, fixed-separation, unsegregated bath. No continuous distribution is fitted. This snapshot also cannot reject such parameters for every cluster or age, or establish a universal BBH distribution. Concentration is consistent with established mass-segregation physics [4], not a newly identified mechanism. Speed fractions in the recorded audit use the matched Plummer escape speed; their discrepancy mixes host-potential and population assumptions and cannot alone diagnose velocity anisotropy or equipartition.

TABLE V. E004 first ballistic returns at identical physical kicks within each model pair. $P_{\rm sh}$ uses 2048 compressed shells; $P_{\rm Pl}$ is the matched Plummer value. Times are Myr and speeds are km s^{-1}.

$\eta_{\rm Pl}$	v_k	$P_{\rm sh}$	$P_{\rm Pl}$	$P_{\rm Pl}/P_{\rm sh}$
0.3	6.8771	0.26250	0.77588	2.9557
0.8	18.3388	0.89636	2.52230	2.8139
0.9	20.6312	1.39333	6.20464	4.4531

C. Frozen spherical potential and identical kicks

Each system is spread over a spherical shell at its snapshot radius. With zero potential at infinity,

$$\Phi_{\rm sh}(r) = -G\left[\frac{M(<r)}{r} + \sum_{r_i \geq r}\frac{m_i}{r_i}\right]. \tag{22}$$

The central value uses the finite limiting shell sum. Exterior mass contributes to potential energy even where its force vanishes. For central launch speed v_k, the energy is $\mathcal{E} = v_k^2/2 + \Phi_{\rm sh}(0)$; the apocenter solves $\Phi_{\rm sh}(r_a) = \mathcal{E}$, and

$$P_{\rm sh} = 2\int_0^{r_a}\frac{dr}{\sqrt{2[\mathcal{E} - \Phi_{\rm sh}(r)]}}. \tag{23}$$

We split the integral at shell boundaries and regularize the turning point. This is a frozen monopole calculation with no drag or encounters.

The snapshot and matched Plummer central escape speeds are 26.547 and 22.924 km s^{-1}. We apply identical physical kicks, chosen as 0.3, 0.8, and 0.9 times the latter speed. Renormalizing by each model's own escape speed would change the physical experiment. All three kicks are bound in both isolated potentials. This binding classification is distinct from tidal removal in CMC [6].

Matching M and r_h does not preserve the central depth or return times (Table V, Fig. 5). Plummer return times are 2.9557, 2.8139, and 4.4531 times the shell values. All exceed the predeclared 10% return-time screen, measured relative to the shell result. This tests transfer of a potential-shape approximation to one state; it does not measure the error of a live-cluster trajectory model.

CMC models relaxation and strong encounters but samples new orbital phases during its Monte Carlo evolution [6]. Connecting snapshot radii is therefore not a continuous physical recoil trajectory. E004 neither follows a displaced remnant nor evaluates its encounters with the measured targets. Those two missing calculations prevent an astrophysical calibration of E003's passage counts.

VII. ANALYTICAL AND NUMERICAL VERIFICATION

For a stationary object, $I(H) = C(H, c) = H$. For nonzero bound kicks, both exposure and occupancy repeat every P, hence $I(nP) = nI(P)$. In the small-kick limit, $I(P) \to \pi$ and $I(P)/P \to 1$.

Near escape, the marginally bound outgoing trajectory obeys $\dot{x} = \sqrt{2}(1 + x^2)^{-1/4}$. The limiting exposure of a full bound excursion is

$$\begin{aligned}\lim_{\eta\to 1^-} I(P) &= \sqrt{2}\int_0^\infty (1+x^2)^{-9/4}dx \\ &= \sqrt{\frac{\pi}{2}}\frac{\Gamma(7/4)}{\Gamma(9/4)} \\ &\simeq 1.0166557018.\end{aligned} \tag{24}$$

Combining this with Eq. (13) gives

$$\frac{I(P)}{P} \sim \frac{\Gamma(7/4)}{\sqrt{\pi}\Gamma(9/4)}(1-\eta^2)^{3/2}. \tag{25}$$

This is a limit from bound orbits. An exactly marginally bound particle does not return in finite time.

The complete verification suite contains 108 passing tests. The original 27 establish the E001/E002 orbital benchmarks. Independent checks integrate the signed equations of motion using DOP853 [11], compare central crossings, monitor energy, and integrate accumulated density as an additional state variable. Core residence is checked with event-segmented trajectories. Other tests cover stationary behavior, sphere thresholds, period accumulation, crossing continuity, density as the derivative of exposure, invalid inputs, and convergence near escape. The strongest near-escape numerical cases reach $\eta = 0.9999$; no claim is made of uniform numerical accuracy arbitrarily close to escape.

For E003, 35 additional checks include independent numerical velocity integrals, analytical stationary and fast-motion limits, and separately integrated DOP853 trajectories with numerical velocity moments. Agreement at 10^{-8} relative tolerance satisfies the predeclared 10^{-6} requirement. Other checks cover units, population/radius/mass scaling, period accounting, and the apocenter screen. Production orbit integrals use absolute and relative tolerances of 10^{-10}.

The remaining 46 tests check snapshot interpretation and shell calculations against direct potential sums, force derivatives, analytical one-shell returns, independent adaptive quadrature, smooth-Plummer recovery, scaling, and tied empirical-CDF jumps. Eight derived snapshot quantities agree with the authors' separately tabulated summary: age, total mass, black-hole mass, system count, object count, binary count, black-hole count, and BBH count. Counts match exactly; relative tolerances are 10^{-7} for masses and 10^{-12} for age. This validates interpretation against another processing of the same state, not an independent astrophysical dataset. Mass columns are already physical; velocities use the N-body time conversion and ages the distinct relaxation-time conversion. External conversion files are parsed without execution.

For E004, approximately equal-mass shell bins use mass-weighted harmonic radii, preserving total mass and central potential. We compare 256, 512, 1024, and 2048 shells. The final two resolutions differ by less than 0.005% in return time and 0.023% in apocenter. Adjacent-resolution differences need not bound error. A direct comparison with all 680110 unique shell radii gives return-time errors of 0.1555%,

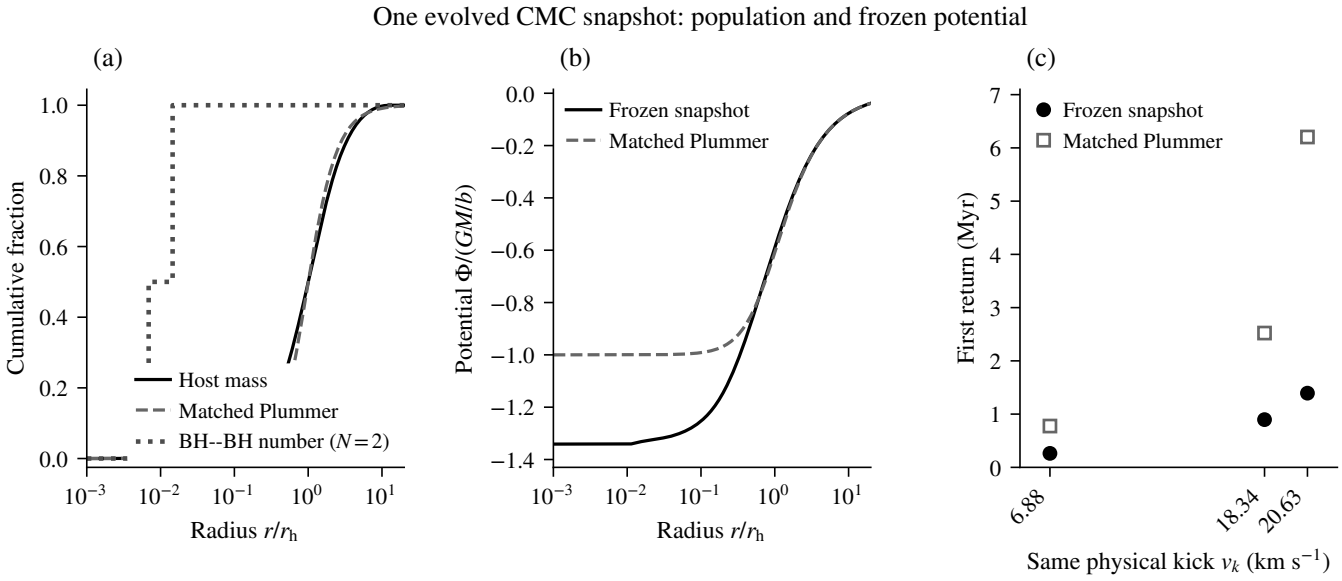


FIG. 5. E004: (a) host mass fractions and the number fraction of the two BBHs; (b) frozen snapshot and matched Plummer potentials; (c) first returns at identical physical kicks. The BBH curve describes two systems, not a fitted companion distribution. Potentials and returns use the 2048-shell representation.

0.00131%, and 0.000568%, respectively, below the 1% numerical screen and far below the potential-shape discrepancies. The finite-shell innermost cavity is force-free; its arbitrarily small-kick limit is not the smooth-core harmonic limit.

Stored-result verification separately reproduces 146 E001/E002 rows, four E003 cases, and E004's five cohort summaries, twelve compressed-shell cases, and 160 plotting points, as well as full-shell and accounting checks. It verifies recorded input and output hashes. Reproduction with the same implementation is distinct from the independent checks above. Grid points and resolution levels are not independent astrophysical observations; solver tolerances are not physical-model uncertainties.

VIII. DISCUSSION AND CONCLUSIONS

A. Claims supported by the combined evidence

The experiments identify distinct approximation errors at successive stages. E001/E002 show how an undamped orbit samples its host: assuming permanent central residence either immediately or after first return does not preserve its exposure history. The wait prescription misses early exposure and eventually overestimates late exposure. The near-escape return time diverges while exposure per bound excursion remains finite. These are properties of the stated collisionless benchmark, not frictional return estimates.

E003 adds the missing velocity kernel. Equation (21) identifies exactly what the density approximation omits, and the selected focusing-dominated cases quantify its effect. The density proxy fails the 10% screen at the two higher kicks while preserving their ordering. Thus density exposure can organize a controlled comparison yet be quantitatively inadequate for passage counts in the same model. The finding is conditional on the given target population, passage radius, masses, and time window; it establishes neither a universal bias nor a calibrated rate.

E004 then tests physical inputs that numerical verification alone cannot justify. The sparse BBH sample does not support a continuous target distribution, and the matched Plummer host fails the return-time screen for this snapshot. The sign of that return discrepancy does not determine the sign of a passage-count correction: concentration, velocities, binary properties, and evolving remnant motion all enter. In particular, it would be unjustified to multiply E003's proxy errors by E004's return ratios. The two comparisons use different hosts and do not supply the joint dynamical calculation needed for that operation.

The combined contribution is consequently an explicit assessment of the approximations between recoil motion and companion accessibility, with reproducible error measurements and identified calibration gaps. It is not evidence that retained remnants reacquire companions at a measured rate. Close passage, exchange or capture, binary hardening, and another merger remain separate physical steps. No formation-channel fraction, observed-event interpretation, or new population discriminator follows from these calculations.

B. Which comparisons would extend the conclusion

Existing collisional cluster models and hierarchical-growth frameworks already contain substantially more physics [2, 3,

6, 10]. The present results motivate a targeted comparison with those methods, not their replacement or a claim that mass segregation and gravitational focusing are new ingredients.

First, companion calibration requires the joint distribution of BBH center-of-mass position and velocity, component masses, semimajor axis, and eccentricity across declared cluster histories. Time weights and object identities must be retained, with whole histories held out for prediction. Repeated snapshots of the same binary are not independent companions, and fitting a radial profile alone does not validate an encounter integral. Sparse regimes should remain explicit rather than being filled by repeated draws from the two available systems.

Second, live recoil tests require resolved collisional motion. Paired kicked and un-kicked runs from the same initial microstate can compare frozen-potential and central-residence approximations against first inward core crossing after exit, finite-window core residence, and cumulative passage counts. Independent microstates and kick directions determine ensemble variation; escape must be distinguished from finite-window non-return. Integration, encounter regularization, and any relaxation-time changes under particle-number scaling require separate checks. These are specified missing comparisons, not executed results of this paper.

Even after these tests, converting access into repeated mergers requires encounter outcomes and binary evolution. We therefore retain a restricted conclusion: ballistic exposure, velocity-weighted passages in an imposed bath, and external potential/population checks can be verified separately, but none substitutes for calibration of their coupled dynamics.

DATA AND CODE AVAILABILITY

The E001/E002 baseline is archived in the version 0.1.1 release [12]; E003 and E004 have immutable result snapshots [13, 14]. The expanded manuscript and portable LaTeX source are maintained in the same public repository. Its paper-specific citation identifies the manuscript revision separately from the baseline software release. The revision expands the synthesis and presentation without changing the recorded experiment outputs.

Python 3.11 is the numerical reference environment, with pinned dependencies. Each experiment manifest records source revisions, source/output hashes, and runtime versions; E004 also pins the external simulation revision and file checksums. The paper manifest binds inputs from all four experiments to the generated tables, vector figures, PDF, and source archive. Project-authored code and documentation use the MIT license. Upstream CMC data and dependencies retain their own rights; raw external files are fetched into verified local cache rather than redistributed. No observational data are used.

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